

SUHAAS CHOUDARY MALLAVARAPU

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SUMMARY

Computer Science sophomore with production deployment experience in full-stack MERN applications, computer vision pipelines, and ML-integrated systems. Shipped a cloud-deployed platform with JWT-secured REST APIs and RBAC, a COLMAP-based 3D reconstruction pipeline, and a CNN classifier achieving 98%+ accuracy. Seeking a Software Engineering internship in backend systems, full-stack development, or AI/ML engineering.

EDUCATION

B.Tech — Computer Science & Engineering — SRM University AP | Aug 2024 – Present

CGPA: 7.4 · DSA, OOP, Operating Systems, DBMS, Computer Networks

Class XII (MPC) — Bhashyam Junior College | 2024

86%

Class X (ICSE) — Loyola Public School | 2022

90%

TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+), C, C++

Frontend: React.js, Tailwind CSS, HTML/CSS, Responsive Design

Backend: Node.js, Express.js, REST API Design, JWT Authentication, RBAC, Middleware

Databases: MongoDB Atlas, Mongoose ODM, PyMongo, NoSQL Schema Design

AI / ML & CV: TensorFlow, Keras, NumPy, OpenCV, CNN Architecture, COLMAP, Photogrammetry, Streamlit

DevOps & Cloud: Git, GitHub, Render, MongoDB Atlas Cloud, Environment Configuration

Core CS: Data Structures & Algorithms, OOP, System Design, DBMS, OS

PROJECTS

EduTrack AI — Full-Stack MERN Student Management Platform *React.js · Node.js · Express.js · MongoDB Atlas ·*

Mongoose · JWT · Tailwind CSS · Render

GitHub: github.com/suhaas1126/EduTrack-AI Live: edutrack-ai-4.onrender.com

- Architected a production-grade MERN application with Admin, Teacher, and Student dashboards, enforcing data isolation via JWT-secured routes and granular RBAC middleware.
- Designed 15+ RESTful API endpoints for student records, attendance, grade management, and reporting, with centralized error handling and structured JSON response conventions.
- Deployed to Render with production environment variable management and MongoDB Atlas cloud integration; built analytics dashboards with AI-assisted academic risk assessment.

3D Reconstruction System — Computer Vision Pipeline *Python · COLMAP · OpenCV · Photogrammetry*

GitHub: github.com/suhaas1126/3D-Reconstruction-COLMAP

- Built a 3D reconstruction pipeline using COLMAP from 300+ real-world images, generating sparse and dense point clouds via Structure-from-Motion (SfM) techniques.
- Implemented feature extraction, feature matching, camera pose estimation, and multi-view stereo workflows to produce accurate 3D scene representations using OpenCV and photogrammetry methods.

Handwritten Digit Classifier — CNN with Real-Time Inference *Python · TensorFlow/Keras · Streamlit · NumPy · OpenCV*

· MNIST

GitHub: github.com/suhaas1126/CodeAlpha_HandwrittenRecognition

- Trained a multi-layer CNN (Conv2D → MaxPooling → Dropout → Dense) on 60K MNIST samples, achieving 98%+ validation accuracy with regularization to minimize overfitting.
- Built a preprocessing pipeline to normalize, reshape, and invert canvas inputs to match MNIST distribution; deployed a real-time Streamlit inference app with per-class confidence scoring.

CERTIFICATIONS & ACTIVITIES

MongoDB Associate Developer (2024) · Verified proficiency in NoSQL data modeling, aggregation pipelines, indexing, and CRUD operations across 5 assessment domains.

Tech Education Channel (50+ Videos) · Produces long-form content on emerging engineering topics; demonstrates technical communication, independent research, and self-directed learning.